

Event-Related Alpha Desynchronization Brain Computer Interface Using Arduino UNO

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Abstract—This report details the hardware architecture and digital signal processing pipeline of a real-time Brain-Computer Interface (BCI) designed to detect Event-Related Desynchronization (ERD) in the Alpha band (8–12 Hz) in the Occipital lobe. By utilizing an isolated electromechanical actuation loop and dynamic statistical thresholding, the system successfully generalizes across tested human subjects with varying baseline neurophysiology, maintaining a robust Signal-to-Noise Ratio while eliminating PWM-induced electrical artifacts.

Index Terms—Brain-Computer Interface, Electroencephalography, Digital Signal Processing, Arduino, Alpha band.

I. INTRODUCTION

THE Alpha rhythm (8–12 Hz) is highly responsive to visual stimuli and attention. The attenuation of this rhythm, known as the Berger effect or Event-Related Desynchronization (ERD), is a well-known phenomenon in biomedical literature. It was first discovered by the German neurologist and psychiatrist Hans Berger in 1929 [1]; in 1934, Edgar Douglas Adrian and Bryan Matthews conducted further studies on the Berger effect, locating it in the Occipital lobe of the scalp [2]; modern literature has described the ERD phenomenon in detail as a gain-control mechanism related to light adaptation [3], [4]. The primary scope of this project is to capture the biological ERD and translate it into a reliable control mechanism for external electromechanical hardware, allowing the user to control a small servomotor by monitoring the Alpha band spectral power. The system combines a compact design with audiovisual feedback, remaining fully operational in both user-operator and user-only setups. Furthermore, the entire BCI was successfully engineered with a total budget of roughly €100.

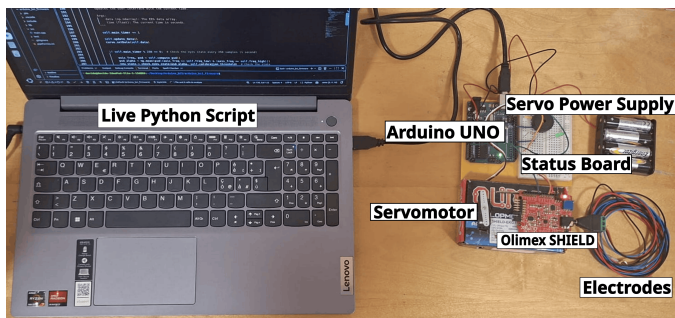


Fig. 1. Brain-Computer Interface setup.

II. HARDWARE ARCHITECTURE & NOISE MITIGATION

To register the Occipital brain activity using electroencephalography (EEG), gold-plated (Au) electrodes have been

used. Au was preferred to Ag/AgCl due to its durability and ease of maintenance, providing a suitable resolution for the experiment and a low impedance combined with conductive paste and a skin cleanser. The three electrodes are placed respectively on Oz (Channel +), Mastoid (Channel -), and Fz (Ground).

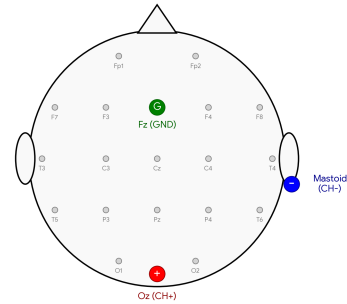


Fig. 2. Electrodes placement.

A significant challenge in closed-loop EEG systems is the acquisition of the signal and its amplification. To mitigate the noise of the EEG system, KONIX EEG Paste was applied to minimize impedance and secure the electrodes on the scalp. To amplify the signal from the usual $\sim \mu V$ range to the correct amplitude for the A/D converter, the electrodes are connected to the OLIMEX SHIELD EKG-EMG biosignal amplifier, with the following pipeline.

Instr. Amp $\rightarrow$ Besselworth 40Hz Lowpass Filter $\rightarrow$ OAMP

The OAMP was left with the default (~ 80) gain, bringing the $G_{\text{total}} = 10 \cdot 80 \cdot 3.46 = 2848$.

The SHIELD is powered by the Arduino board at 5 V and sends the digital data to the A0 port of the Arduino. The board also powers a green LED and a piezo speaker to provide visual and audio feedback to the user about the state of the machine. The EEG data is then sent in a 4 byte packet containing a start and an end sequence and the 10 bit signal in between. This configuration allows the computer program to correctly recognize the single data packet and synchronize with the board. A USB cable both powers the system and allows communication between the board and the computer.

The servomotor is powered separately with a 6 V battery to avoid PWM noise in the SHIELD: if the squared wave sent from Arduino to power the servo is connected in parallel with the board, a small percentage of the signal enters the amplification pipeline, producing huge bumps that completely disrupt the signal.

III. BCI PIPELINE

The entire BCI pipeline is shown below.

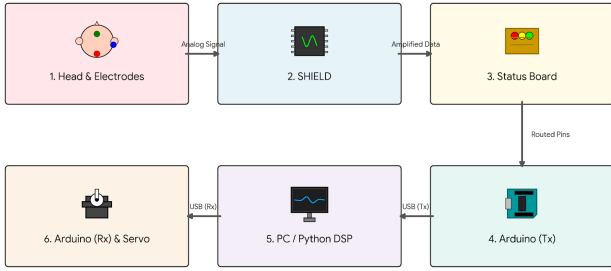


Fig. 3. The BCI Pipeline.

- 1) The electrodes placed on the head record the biopotential. Common mode signal is mitigated by the Fz reference electrode, and the Mastoid CH- is introduced as a differential input to subtract common muscle artifacts during the recording.
- 2) The SHIELD amplifies and filters the analog signal.
- 3) The board shows the BCI status. A green LED is on during all the data acquisition procedure.
- 4) The Arduino UNO converts the data to a digital 4-byte packet. Data is converted based on the 5 V scale based on the relation

$$\text{Data} = \frac{S}{V} \cdot 2^{10}$$

Where S is the analog signal and V is the power line of the board (5 V).

- 5) The PC reads the packet and analyzes the data. The processing is described in Section IV.
- 6) The PC feedback is sent back to the Arduino as a ASCII conversion of a char value, which is evaluated to correctly actuate the servo.

IV. SIGNAL PROCESSING

The continuous data stream is sampled at 250 Hz to provide a solid and precise signal. The digital signal processing is managed by the Python live script. The script is responsible for calibration, synchronization of the data collection process, data plotting, data filtering, spectral power estimation and response evaluation, and data storage. The choice of the coding language is justified by the useful libraries for signal processing and the ease of use for running live scripts.

A. Data collection

The script is set to accept data from a specified serial port; the packet received is then checked with the start and end sequence and the information content is then stored in a numerical array. The buffer records 5 minutes of data ($5 \cdot 60 \cdot F_s = 75 \cdot 10^3$ samples), providing a sufficient time window without the risk of RAM overflow. After exactly five minutes from the start of the experiment, the array is stored in a CSV file to allow for offline analysis, the script keeps running until stopped by the user.

B. Bandpass Filtering

A 4th-order Butterworth bandpass filter isolates the 8–12 Hz frequency band. Choosing to use Infinite Impulse Response (IIR) Filters allows low-latency filtering due to the low order, and a good mitigation of cutoff frequencies with a sufficient small ripple effect for the specific application.

C. Spectral Estimation and Windowing

Power Spectral Density (PSD) is calculated using Welch's method. The estimation is made on a 4 second window during calibration phase, and is then executed on every new second (250 samples) of data. Such time windows provide robust threshold estimation and dynamic power evaluation to correctly describe the power increase/decrease.

V. DYNAMIC CALIBRATION & RESULTS

Because biological signals exhibit immense physiological variance across individuals, static amplitude thresholds frequently fail. The system provides high adaptability to different users by implementing an initial phase of calibration. The process is divided into 4 steps:

- 1) 5 second recording in active (eyes open) state. Estimation of low Alpha-band mean power m_{active} .
- 2) 5 second recording in rest (eyes closed) state. Estimation of high Alpha-band mean power m_{rest} .
- 3) Evaluation of spectral power levels. The script returns an error if the high level is below the lower level and prints a warning if the high level is less than 1.5 times the low level (which might be caused by bad electrode connection).
- 4) Estimation of threshold T .

A. Artifact Rejection & Outliers Robustness

To avoid including any movement that the user might do during the change from active (eyes open) and the rest (eyes closed) state, a transition time is added between the calibration steps. For additional mitigation of accidental artifacts, the Spectral Power estimation is calculated over a 4 second window extracted from the 5 second window registered, so the first second of the calibration procedure is automatically rejected.

To mitigate servo twitching during the experiment—a behavior shown in early tests that was related to false positives/negatives and only lasted one window frame—the state evaluation system compares the new power window P_{new} to the last two windows evaluated P_{current} and P_{past} . The algorithm logic is:

- if $P_{\text{new}} == P_{\text{current}}$ AND $P_{\text{new}} \neq P_{\text{past}}$: the servo changes its state.
- else: the servo does not change its state.

Although this method introduces 1 second of latency to the actuation of the mechanism, it provides a much more precise response, actuating correctly in response to a change of state.

B. Statistical Thresholding

The transition threshold T is defined dynamically using a 3-sigma confidence interval based on the user's active (eyes open) Alpha power:

$$T = m_{\text{active}} + 3 \cdot s_{\text{active}}$$

Where m_{active} and s_{active} are the sample's mean and standard deviation during the active phase (eyes open).

This threshold provides precise response to the user's input, as shown in the test trials.

C. Offline Validation

Offline MATLAB analysis of continuous Alpha power revealed significant physiological variance among the tested subjects. Subjects demonstrating high Event-Related Desynchronization achieved a linear Signal-to-Noise Ratio (SNR) exceeding 4.0 (with a registered maximum of 6.69 dB). However, the system also successfully adapted to subjects with heavily attenuated Alpha responses (SNR 3.15/4.98 dB). A static threshold calibrated for high-amplitude subjects would have resulted in a false-negative failure rate for low-amplitude users. The implementation of the dynamic 3-sigma statistical threshold successfully mitigated this physiological variance, ensuring robust electromechanical actuation across all subjects.

To demonstrate adaptability, high and low ERD samples are shown in Table I and Figures 4–5. Subject 1 shows an exceptional example of ERD, while Subject 2 presents a milder, yet noticeable difference. It should also be noted that some of the subjects presented anomalous spikes that were recorded in the low-power region, but most of these outliers were successfully tackled by the two-windows delay implemented in the algorithm.

TABLE I
SIGNAL-TO-NOISE RATIO (SNR) ANALYSIS OF ALPHA-BAND
EVENT-RELATED DESYNCHRONIZATION

Subject	Baseline (Opened)	Active (Closed)	SNR (Linear)	SNR (dB)
Subject 1	0.3261	1.5218	4.67	6.69
Subject 2	0.2042	0.6426	3.15	4.98

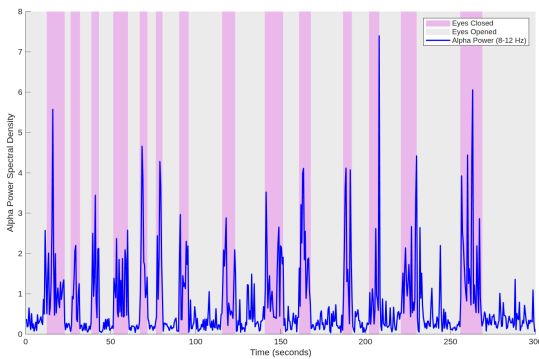


Fig. 4. Subject 1 (6.69 dB SNR)

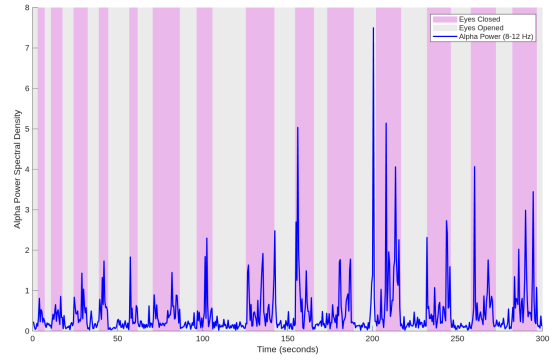


Fig. 5. Subject 2 (4.98 dB SNR)

The cases show a correct registration of Alpha-band spikes during resting state (colored windows). The dynamic threshold allows the servo to be controlled adapting to the subject's physiological PSD. Preliminary testing suggested approximately 95% correct actuation, with an average response delay of about 1.5–2 s, trading response speed for improved robustness over false positives/negatives.

VI. DISCUSSION & LESSONS LEARNED

Hardware-software co-design is critical in biomedical engineering. Even with correct electrode placement, DSP algorithms cannot work properly with a physically compromised signal. For this reason, securing the hardware grounding was a mandatory prerequisite for software filters and algorithms to function. Furthermore, the necessity of adaptive baselining was proven: static thresholds are unviable for general-purpose BCIs, providing good suitability for the majority of subjects.

VII. CONCLUSION

By combining strict hardware isolation with an adaptive, statistically grounded software threshold, this system provides a robust framework for real-time BCI control, generalizable across multiple users without manual recalibration.

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